



Merchandise Store

Funnel & Conversion Analysis

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The Main Question

Where do users drop off in the path from landing on the site to completing a purchase, and what does that mean for the future of the business and how it should prioritize its marketing, products, and retention efforts.

Data Source

Google's official GA4 sample e-commerce dataset (BigQuery public data) for the period of November 1st 2020 to January 31st 2021 containing real traffic from the Google Merchandise Store.

Data Analysis Process:

1. Queried raw event data directly in BigQuery using SQL queries to isolate funnel stages, channel performance, device performance, landing page entries, daily/weekly trends, cart abandonment, and new vs. returning visitor behavior.
2. Cleaned and reshaped results in Excel and Power Query (unpivoting wide tables, extracting/cleaning landing page labels, calculating weekday aggregates).
3. Built supporting calculations in DAX (e.g., overall conversion rate as a measure).
4. Designed and built an interactive two-page dashboard in Power BI, using a consistent, color system: blue for neutral metrics, green reserved for positive standouts, and red used for the metrics that represents a real business problem (cart abandonment).

Key Findings

- The funnel leaks early, not at checkout. Only 22.9% of sessions view products, and just 20.5% of viewers add items to cart. However, once a user starts checkout, nearly half (45.5%) complete the purchase. The opportunity for change is driving product engagement, not fixing checkout friction.
- Referral traffic converts best (1.89%), paid search converts worst (0.98%). Despite presumably costing money to acquire, paid search underperforms organic, direct, and referral traffic on conversion quality.
- Returning visitors convert at nearly 7x the rate of new visitors (9.26% vs. 1.36%), despite making up under 4% of total sessions — a strong signal that retention/remarketing investment may outperform additional top-of-funnel acquisition spend.
- Mobile converts slightly better than desktop (1.71% vs. 1.62%), countering the common assumption that mobile underperforms.
- Clear seasonal pattern. Conversion peaks on Cyber Monday (3.22%), stays elevated through early-to-mid December, dips around Christmas, and drops again in early January before a partial late-January recovery.
- Friday is the strongest day for purchase volume (814 total purchases across the period), with Sunday the weakest.

Recommendation

Prioritize engagement on the page including better merchandising, recommendations, or content on category/product pages) and returning-visitor retention such as email and marketing over checkout optimization. The data shows the biggest, most addressable leak is early in the funnel, and the highest-value segment (returning visitors) is currently a small share of total traffic.

Limitations

- Google replaces some field values with placeholders such as “others” and “data deleted” these were identified and excluded from channel-level conclusions rather than treated as real categories.
- Channel and device segment totals can exceed total session counts, since a single user may appear in multiple channels/devices across separate visits within the period.